

T-SF32-Display_V1.0 Power consumption test

Date: 2026_06_26
Test Board: T-SF32-Display_V1.0 && T-SF32-Keyboard_V1.0
Software: menu_app_v1.0
Test organization: LILYGO

1. Test Environment Description

1.1 Test environment(测试环境)

Test Tools	Sampling speed	Test Methodology	Input voltage	Test environment temperature
BLU939	100000 times/s	Battery Interface Measurement	+4.0V	25°C/77°F

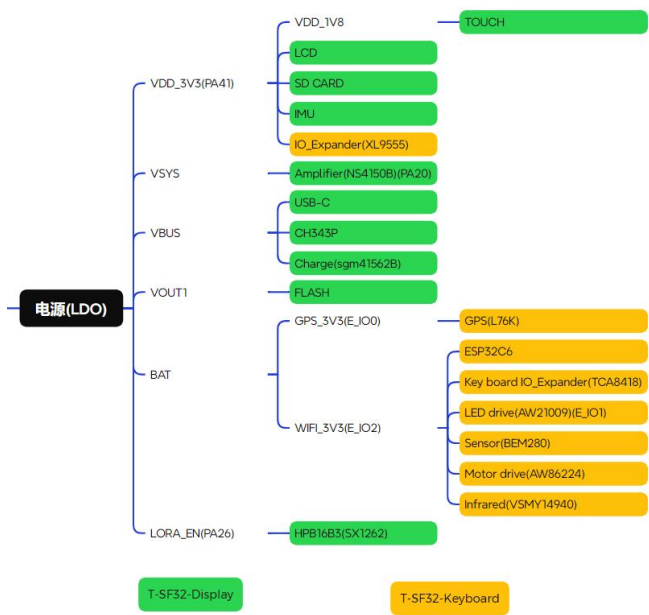
1.2 Test device model(测试器件型号)

1.2.1 T-SF32-Display

Test deviece	Screen	Lora	BlueTooth	Loudspea ker	IMU	TF
Component model	H0216F002AM	HPB16B3(SX1262)	SIFLI	NS4150B	BHI260AP	MicroSD

1.2.2 T-SF32-Keyboard

Test deviece	Slave MCU	Keyboard	Key led	GPS	Sensor	Motor	IR
Component model	ESP32C6	TCA8418	AW21009QNR	L76K	BME280	AW86224AFCR	VSMY14940



Power Control Diagram

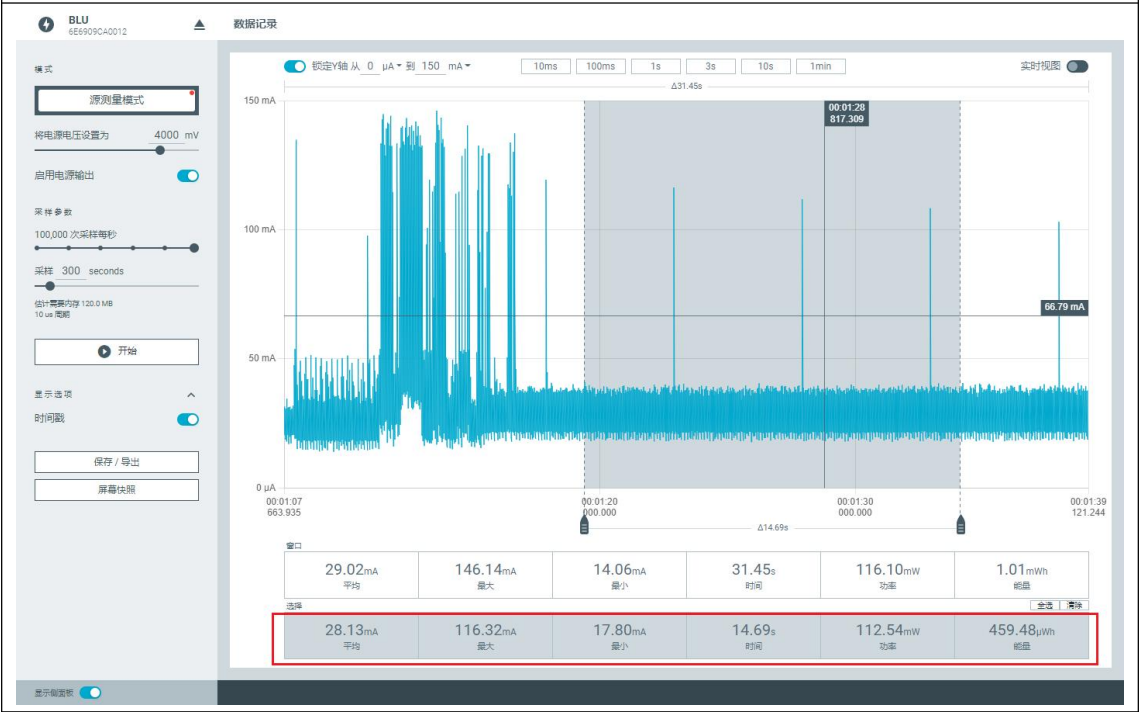
2. Test Physical Object Image

3. Power Consumption Test

- The screen is set to 100 brightness level when turned on.
- LORA transmission is the maximum transmission power, while LORA reception is in the enhanced reception mode.
- The audio playback volume is set at 50% level.
- This firmware uses the STA mode for the ESP32C6.

SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 28.13mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	SLEEP	ON
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	ON	SLEEP	ON
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	ON	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 32.04mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	TX	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 47.62mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	RX	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	OFF	ON	ON	SLEEP	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 22.99mA



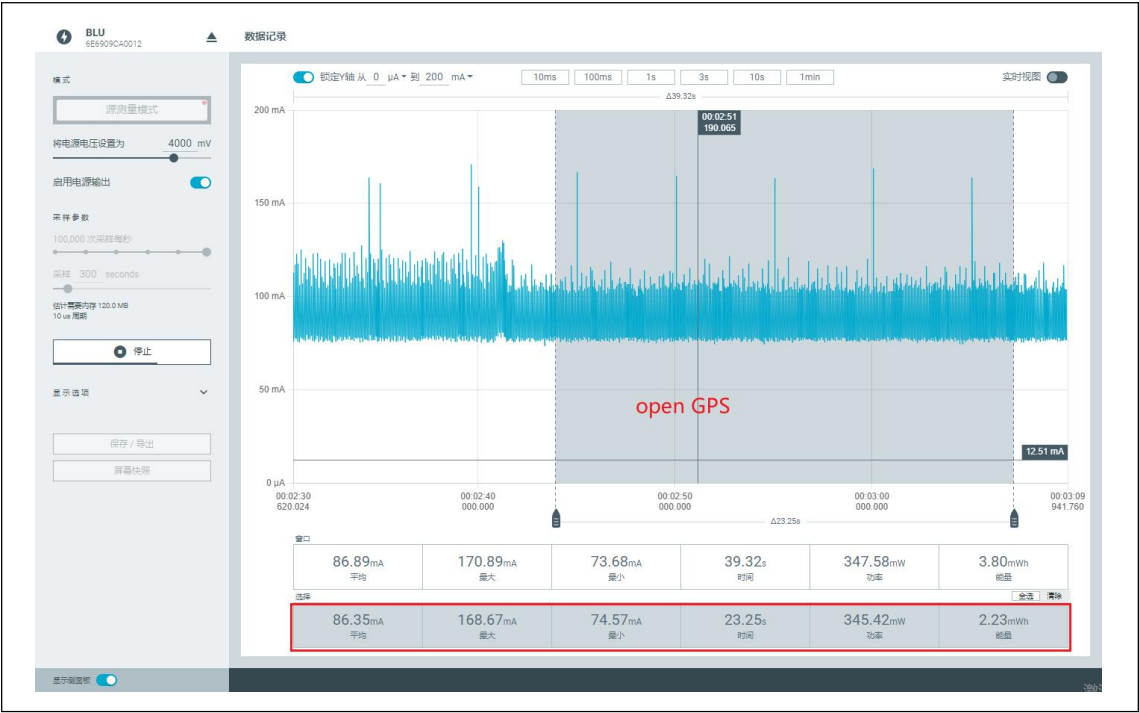
SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
ON	OFF	ON	ON	ON	OFF	OFF	

Test result: average 91.60mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	ON	OFF	OFF	OFF	OFF	OFF	

Test result: average 86.35mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Active	ON	ON	ON	SLEEP	OFF	SLEEP	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	ON	OFF	

Test result: average 193.08mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
HIBERNATE	OFF	OFF	OFF	OFF	OFF	OFF	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 95.44mA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
HIBERNATE	OFF	OFF	OFF	OFF	OFF	OFF	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 4.8uA



SFLB52X	LCD	TOUCH	SD CARD	IMU	NS4150B	LORA	BLUETOOTH
Shutdown	OFF	OFF	OFF	OFF	OFF	OFF	OFF
WIFI	GPS	KEYBOARD	KEY_LED	SENSOR	MOTOR	INFRARED	
OFF	OFF	OFF	OFF	OFF	OFF	OFF	

Test result: average 0.87uA

